



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 38 summarizes the current fairness state for upload 38. Accuracy changed to +0.787 and fairness score changed to +0.954. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.887.

ACCURACY

0.787

▲ +0.787

FAIRNESS SCORE

0.954

▲ +0.954

COMBINED SCORE

0.887

ThresholdOptimizer_mitigated_threshold

DISPARATE IMPACT

0.820

Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	compas_dt_baseline.pkl	comparison

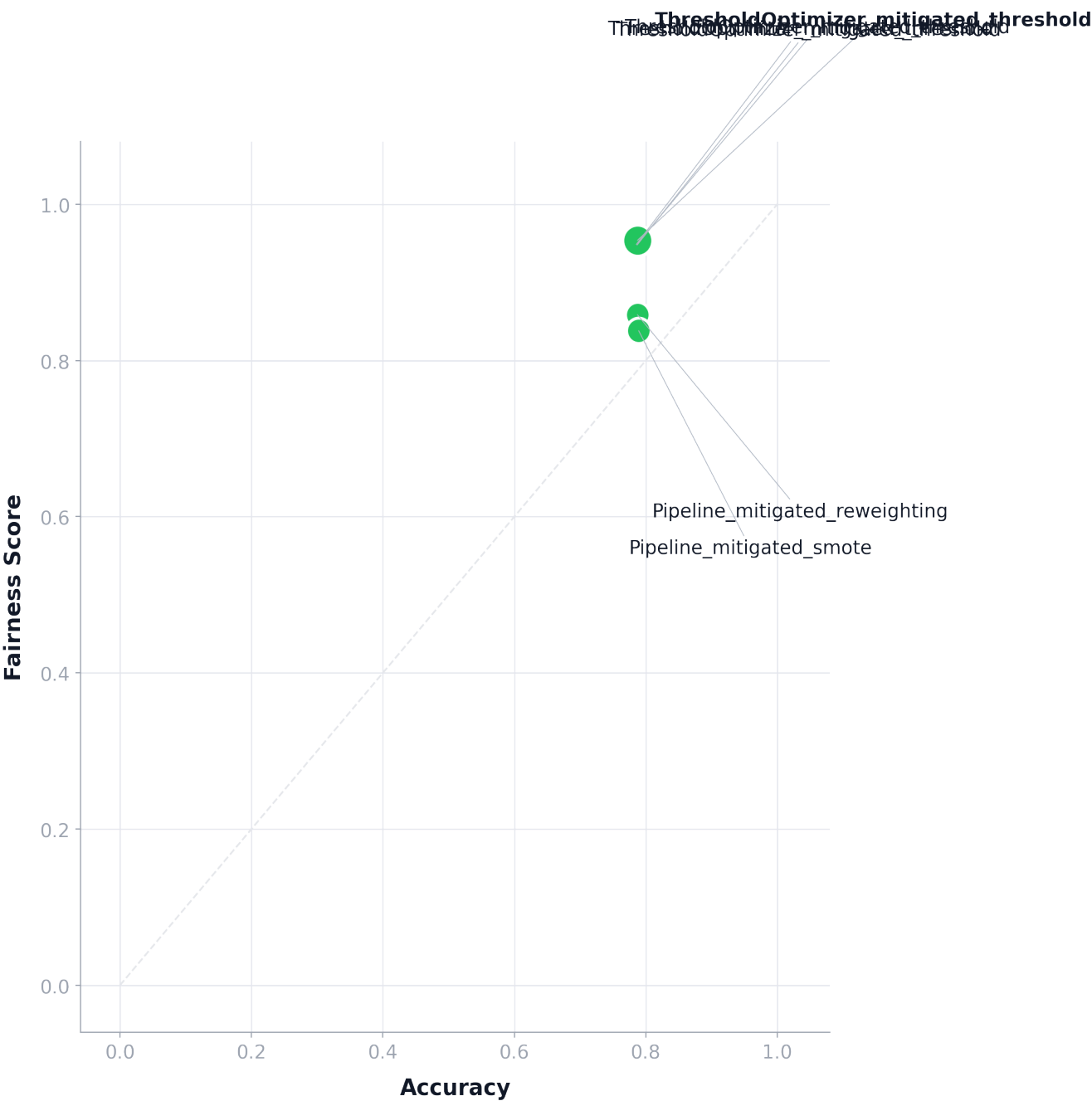
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.081	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.012	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.820	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.787	0.954	0.081	0.012	0.820	0.887
★ ThresholdOptimizer_mitigated_threshold	0.788	0.950	0.084	0.016	0.813	0.885
★ ThresholdOptimizer_mitigated_threshold	0.786	0.948	0.083	0.021	0.816	0.883
★ ThresholdOptimizer_mitigated_threshold	0.787	0.948	0.084	0.021	0.813	0.883
Pipeline_mitigated_reweighting	0.787	0.858	0.166	0.117	0.633	0.830
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Pipeline_mitigated_reweighting	0.787	0.858	0.166	0.117	0.633	0.830
Pipeline_mitigated_smote	0.789	0.838	0.185	0.139	0.609	0.818
Pipeline_mitigated_smote	0.789	0.838	0.185	0.139	0.609	0.818
Pipeline_mitigated_smote	0.789	0.838	0.185	0.139	0.609	0.818

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.